Prabhavit Gupta

Senior Fullstack Engineer | AI Systems, Model Context Protocol (MCP), Agents & Backend Infrastructure

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Executive Summary

- **Production Agent Architectures & Model Context Protocol (MCP):** Software engineer with 3 years commercial experience architecting and shipping production-grade AI systems, autonomous agents, and tool-calling infrastructure. Deep practical experience building custom Model Context Protocol (MCP Server & Client) integrations and multi-step agent workflows.
- **Fullstack Backend, Microservices & Data Infrastructure:** Robust fullstack engineering expertise across Python (FastAPI, AsyncIO, PyTest), TypeScript, and PostgreSQL. Experienced designing resilient microservices, distributed background task queues, and cloud infrastructure on Microsoft Azure and Docker.
- **Observability, Evals & Startup Velocity:** Experienced in high-velocity startup environments, establishing rigorous evaluation harnesses (RAGAS / automated benchmark evals), LLM tracing (Langfuse / OpenTelemetry), context/memory management for long-running agents, and security-first enterprise API integrations.

Technical Skills

- **Agentic AI & Tool Protocols:** Model Context Protocol (MCP Server & Client), LangChain, LangGraph, Deep Agents, ReAct Planning, Tool Calling, Stateful Multi-Turn Memory, Context Window Optimization, Human-in-the-Loop.
- **LLMs, Voice AI & Evals:** OpenAI GPT-4o, Claude 3.5 Sonnet, Langfuse Tracing, Prompt Engineering, Structured Outputs (Pydantic, JSON Schema), Audio/Speech Pipelines (Whisper STT/TTS), RAGAS Automated Evals.
- **Backend & Microservices:** Python (Advanced - AsyncIO, FastAPI, PyTest, Typing), TypeScript, Node.js, PostgreSQL (pgvector), Redis (Caching & Queues), REST APIs, gRPC, Microservice Architectures.
- **Cloud, DevOps & Security:** Microsoft Azure, Docker, CI/CD (GitHub Actions), Linux, OpenTelemetry Tracing, Structured Logging, OAuth2 / Secure Credential Isolation, Penetration & Contract Testing.
- **Engineering Methodologies:** Startup Product Engineering, Systems Architecture, Clean Code (SOLID), Agile Delivery, High-Traffic Reliability.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and shipped production AI platforms and autonomous agent systems for European startups and tech scale-ups, managing end-to-end delivery from prototype to production.
- Built custom Model Context Protocol (MCP) servers and clients, standardizing tool-calling interfaces to securely connect LLM agents to Postgres databases, external REST APIs, and third-party SaaS services.
- Designed multi-turn agent orchestration layers with persistent state management, dynamic context trimming, and human escalation gates using LangChain and LangGraph.
- Implemented real-time LLM observability and distributed tracing using Langfuse and OpenTelemetry, monitoring latency, token economics, and hallucination rates in live environments.
- Developed automated evaluation suites and regression test harnesses, systematically tracking model output quality, precision, and tool-invocation accuracy across prompt releases.
- Deployed containerized microservices on Microsoft Azure and Linux infrastructure, implementing robust CI/CD pipelines and security-first credential management.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Built scalable backend microservices in Python and TypeScript for automated document ingestion, classification, and real-time audio transcription processing.
- Developed vector indexing pipelines and caching layers with PostgreSQL and Redis, ensuring sub-second query response times across high-volume data workloads.
- Created comprehensive unit, contract, and integration test suites using PyTest, maintaining high code quality and test coverage across CI/CD deployments.
- Collaborated with product and engineering founders to rapidly translate business objectives into validated, scalable software architectures.

Key Engineering Projects

- **Autonomous Tool-Using Agent Platform (LangGraph + Python + MCP + Postgres):** Multi-agent collaborative execution system integrating external tools via Model Context Protocol, featuring state persistence,

Langfuse tracing, and human-in-the-loop validation.

- **Real-Time Voice & Document Intelligence Engine (FastAPI + Whisper + pgvector):** Microservice architecture combining speech-to-text audio processing, dense semantic retrieval, and structured JSON output extraction with automated validation guardrails.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, decision systems, computational theory, and complex systems modeling.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the European Union without sponsorship; available immediately for remote engagement.